DAT101: Mandatory assignment 3.2

Introduction

Loop the Loop: Mastering JavaScript Iteration!

Get ready to embark on a thrilling journey into the heart of JavaScript loops! In this assignment, we'll unravel the power of repetition and control flow, exploring how `for`, `while`, and `do` loops can automate tasks, create dynamic patterns, and bring your code to life. You'll tackle practical problems, engage in stimulating discussions, and emerge with a deeper understanding of how loops are the backbone of interactive web experiences..

To get started with this JavaScript assignment:

1. **Locate Your Files:** Navigate to the Git repository you created in Assignment 1.1. You'll find two files:
    - `task_3-2.html`
    - `task_3-2.mjs`
2. **Use the Provided Files:** These are the foundation for your work. Do not rename them.
3. **HTML for Collaboration:** The `task_3-2.html` file is where you'll document the results of your group discussions.
4. **MJS for JavaScript Code:** Write your JavaScript code within the `task_3-2.mjs` file.
5. **Read Carefully:** Review the assignment descriptions thoroughly. Pay close attention to the details to ensure you include all required elements in your solution.
6. **Meaningful Variable Names:** Always use descriptive English variable names that clearly convey their purpose.

Resources:

- W3Schools for loops: https://www.w3schools.com/jS/js_loop_for.asp
- MDNfor loops: https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Statements/for
- W3Schools do/while: https://www.w3schools.com/jsref/jsref_dowhile.asp
- MDN do/while: https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Statements/do...while
- W3Schools while: https://www.w3schools.com/js/js_loop_while.asp
- MDN while: https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Statements/while
- MDN loops: https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide/Loops_and_iteration

Learning Objectives:

Upon successful completion of this assignment, you will demonstrate the ability to:

- The use of "for" loops in programming.
- The use of "while" loops in programming.
- The use of "do" loops in programming.

Learning Outcomes:

Upon successful completion of this assignment, you will demonstrate the ability to:

- Explain how a "for" loop works and when we use it in programming, including the use of code blocks within "for" loops in JavaScript.
- Explain how "do" and "while" loops work, how they differ, and how they are structured in JavaScript.
- Utilise code blocks within "do" and "while" loops in JavaScript.
- Explain how all the conditions in a "for" loop function, and be able to determine how many iterations a complete loop will execute.
- Use comparison operators as conditions within loops in JavaScript.
- Recognize endless loops.
- The ability to recognize endless loops.

Collaborative Discussion

Looping Together, Learning Individually

This activity is designed to promote collaborative learning and deepen your understanding of the fundamental principles of loop constructs covered in Module 3. Engage in group discussions to explore the questions below. While collaboration is encouraged, your final submission should be an individual reflection that demonstrates your own understanding and insights. Feel free to leverage AI tools to enhance your analysis and synthesis of the discussion points.

Discuss the following questions in your group, and write down your discussion in the bottom part of the HTML document before the JavaScript element:

Discussion Questions:

1. In part 1, is it possible to solve the problem using a single "for" loop?
2. In part 2, can we create what is known as an "infinite" loop within the condition of the "while" loop? Describe what would or would not be necessary to achieve this. Provide a thorough explanation of your answer.
3. Discuss briefly how quickly or slowly a computer might be able to compute this, considering the number of guesses and the time it takes. How does the program behave while it is guessing? Can the program perform other tasks concurrently?
4. Explain the relationship between rows and columns in nested "for" loops. Can we, for instance, change the order, considering rows to be innermost or columns to be innermost, or does this not matter?

Individual Submission: After your group discussion, take some time to individually reflect on the key takeaways and insights you gained. Use this information to inform your final submission, ensuring it accurately reflects your own understanding.

Remember: The goal is to learn from each other and develop your own understanding. Feel free to share your insights and experiences, but be sure to formulate your own responses for submission.

Individual Tasks (Collaborative Learning Opportunities)

Feel free to engage in group discussions and brainstorm ideas, but remember that each student is responsible for completing and submitting their own unique solutions for each part of the assignment

- ### Part 1 Counting Up and Down (8 points)

Use "for" loops to generate two lines on the HTML page. One should count from 1 to 10, and the other should count from 10 to 1. Use only two lines to print the rows.

- ### Number Guessing Game (12 points)

Create a program that can guess a number between 1 and 60. Declare a variable and assign it a value, for example, 45. Let the computer "guess" by generating a random number. Use a "while" loop and the "random" function. Keep the "while" loop running as long as the "guessed number" is incorrect. Print the number once the "while" loop has completed. You do not need to print anything while the "while" loop is in progress.

- ### Guessing Game - Level Up! (20 points)

Take the program from part 2 and expand it to guess a number between 1 and one million. Print the number of guesses as well as the number of milliseconds it took to guess the number. HINT: Use the `Date.now()` function to measure time.

- ### Part 4 Prime Number Hunter (15 points)

Use a "for" loop and a "while" loop to find all prime numbers greater than 1 and less than 200.

    - *HINT: A prime number is any natural number greater than 1 that is only divisible by itself and 1. The number 1 is not a prime. (See Wikipedia on primes or ask your AI).*

- ### Part 5 Nested Loops & Patterns (15 points)

Create two loops that print 9 columns and 7 rows with the text "K1, R1" for the first cell, "K2, R1" for the second cell, and so on.

    - *Hint: Use what we call nested loops. This is a "for" loop within another "for" loop.*

Use the provided `printOut` function to print each row with its sets of columns; remember to place this in the right level of the nested for loops. The output should look like this:

K1R1 K2R1 K3R1 K4R1 K5R1 K6R1 K7R1 K8R1 K9R1
K1R2 K2R2 K3R2 K4R2 K5R2 K6R2 K7R2 K8R2 K9R2
K1R3 K2R3 K3R3 K4R3 K5R3 K6R3 K7R3 K8R3 K9R3
K1R4 K2R4 K3R4 K4R4 K5R4 K6R4 K7R4 K8R4 K9R4
K1R5 K2R5 K3R5 K4R5 K5R5 K6R5 K7R5 K8R5 K9R5
K1R6 K2R6 K3R6 K4R6 K5R6 K6R6 K7R6 K8R6 K9R6
K1R7 K2R7 K3R7 K4R7 K5R7 K6R7 K7R7 K8R7 K9R7

- ## Grade Simulator & Sorting Challenge (10 points + Bonus Hint)

Simulate 5 student grades using the `Math.random()` function, from 1 to 236 (inclusive).

For each grade, print the student's grade (A to F) based on the point distribution provided:

  - A: 89% – 100%
  - B: 77% – 88%
  - C: 65% – 76%
  - D: 53% – 64%
  - E: 41% – 52%
  - F: 0% – 40%

- ## Part 7 Dice Rolling Extravaganza (20 points)

Simulate 6 dice and print how many "throws" it takes to get:

- 1 2 3 4 5 6 (full straight)
- 3 pairs
- 2 of a kind and 4 of a kind (tower)
- All the same (Yahtzee)

Submission:

Submit your commit URL on Canvas LMS

How to Submit:

To submit your work for Task 3.2 using VS Code, follow these steps:

1. **Prepare and Push Your Changes**
   a. Save the changes you made to the `task_3-2.html` and `task_3-2.mjs` file within your Git repository.
   b. **Commit** these changes, and then **Push** them to your remote GitHub repository.
2. **Obtain the Commit URL**
   a. Navigate to your GitHub repository in your web browser.
   b. Locate the specific folder of your Task 3.2 files that you want to submit.
   c. **Copy the URL** from your browser's address bar.
3. **Final Submission**
   a. Paste the copied **URL** into the Canvas submission text field.

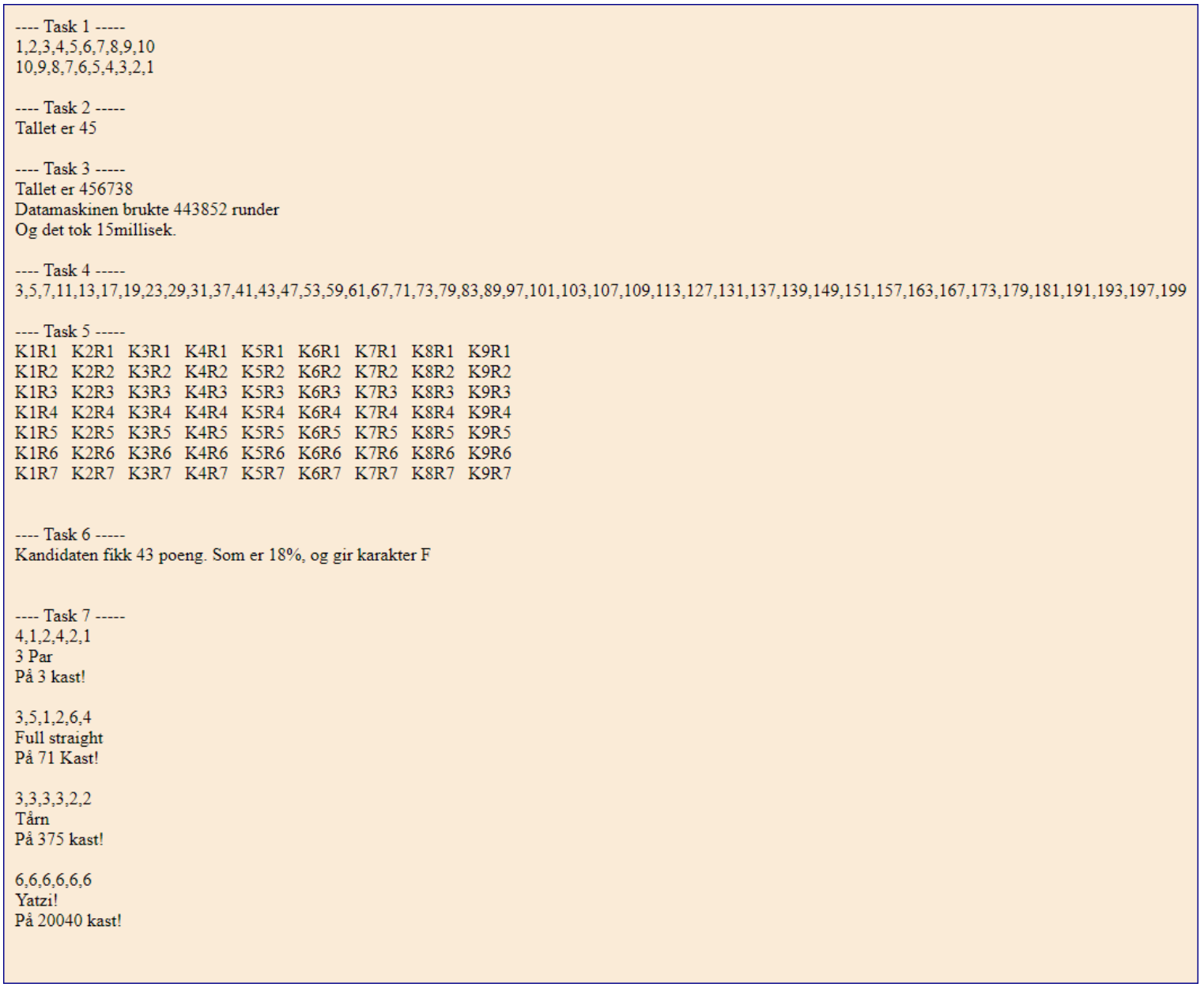
Additional Tips

- **Version Control is Key:** Use Git to track changes and easily revert to earlier versions if needed. This is a vital skill for developers.
- **Modules for Organization:** The `.mjs` extension means this is a JavaScript module. Modules help structure code and avoid naming conflicts. We'll explore them further in the course.
- **Ask Questions:** If you have any questions about the assignment, Git, or modules, don't hesitate to ask me or your fellow students. You can find us on the course Discord server.

- **HTML output:** See the attached image for an example of how your HTML output might look. You can choose your preferred language, but remember that code syntax and variable names must be in English.

Best of luck! I'm confident you'll do great on this assignment 😉.

Sincerely,
Arne-Thomas (Lecturer)



---- Task 1 -----
1,2,3,4,5,6,7,8,9,10
10,9,8,7,6,5,4,3,2,1

---- Task 2 -----
Tallet er 45

---- Task 3 -----
Tallet er 456738
Datamaskinen brukte 443852 runder
Og det tok 15millisek.

---- Task 4 -----
3,5,7,11,13,17,19,23,29,31,37,41,43,47,53,59,61,67,71,73,79,83,89,97,101,103,107,109,113,127,131,137,139,149,151,157,163,167,173,179,181,191,193,197,199

---- Task 5 -----
K1R1 K2R1 K3R1 K4R1 K5R1 K6R1 K7R1 K8R1 K9R1
K1R2 K2R2 K3R2 K4R2 K5R2 K6R2 K7R2 K8R2 K9R2
K1R3 K2R3 K3R3 K4R3 K5R3 K6R3 K7R3 K8R3 K9R3
K1R4 K2R4 K3R4 K4R4 K5R4 K6R4 K7R4 K8R4 K9R4
K1R5 K2R5 K3R5 K4R5 K5R5 K6R5 K7R5 K8R5 K9R5
K1R6 K2R6 K3R6 K4R6 K5R6 K6R6 K7R6 K8R6 K9R6
K1R7 K2R7 K3R7 K4R7 K5R7 K6R7 K7R7 K8R7 K9R7

---- Task 6 -----
Kandidaten fikk 43 poeng. Som er 18%, og gir karakter F

---- Task 7 -----
4,1,2,4,2,1
3 Par
På 3 kast!

3,5,1,2,6,4
Full straight
På 71 Kast!

3,3,3,3,2,2
Tårn
På 375 kast!

6,6,6,6,6,6
Yatzi!
På 20040 kast!